Introduction Waves

1 Summary: Oscillations

By now, we know that an **oscillation** is a repeated back-and-forth motion around an equilibrium position.

We have learned that every oscillation can be described by a few important quantities:

- **Amplitude** (A): the maximum distance from the equilibrium position.
- **Period** (T): the time for one complete oscillation.
- **Frequency** (f): the number of oscillations per second.
- The relationship
$$f = \frac{1}{T}.$$

We also discovered that many oscillations can be described by a **sine** or **cosine** function. The shape of the graph changes when we change the amplitude, frequency or phase.

> What have we learned?
>
> An oscillator moves back and forth around an equilibrium position.
> Different oscillators may look different, but they can all be described using the same mathematical ideas.

We were looking at spring and simple pendulum and described their oscillations with the same mathematical expression.

2 From Oscillations to Waves

Introduction: What is actually moving?

Experiment: Hold up a slinky (or a long rope) and send a single pulse through it. Do not explain anything yet.

Questions for the students

- What just moved?
- Did this end of the slinky travel all the way to the other side?
- Did any single part of the spring move from one end to the other?

Time for discussion.

> *"If the spring itself did not travel across the room, how did the movement get to the other side?"*

Conclusion:
Something travelled from one end to the other, but it was **not the material**.

A wave is a **travelling disturbance** that transfers energy through many oscillators. Those oscillators are connected, we call this coupling. A disturbance that is travelling through the medium is called **wave**.

Today we are going to answer a new question:

How does an oscillation become a wave?

Examples - in groups

Can you think of an example for a wave? What are the oscillators and how are they connected

Wave	What is oscillating?	What is the coupling?
Wave on a rope	Small pieces of the rope	The tension in the rope pulls neighbouring pieces.
Water wave	Water particles move up and down (or in small circles)	Forces between neighbouring water molecules (hydrogen bonds).
Sound wave in air	Air molecules oscillate back and forth	Collisions and pressure forces between neighbouring air molecules.
Slinky wave	Each coil of the slinky	The spring force between neighbouring coils.
Stadium wave (Mexican wave)	People stand up and sit down	Each person reacts to their neighbours.
Domino chain (analogy)	Each domino rotates once	Each domino knocks over the next one.
String of a guitar	Every tiny piece of the string	The tension in the string.

Table 1: Examples of waves, their oscillators, and the coupling between them.

Activity: Become a Wave

In this activity, we want to create waves ourselves and discover two different kinds of waves

Part 1: Transverse Wave

Ask all students to stand in a line.

Explain the following rules:

- Each student represents one oscillator.

- Stay where you are throughout the activity.
- Watch the person on your left.
- When they raise their hands, wait about one second.
- Then raise your own hands.
- Lower them again after one second.

Start with the first student. A wave should travel through the line.

Ask the students:

- Did anyone move across the room?
- In which direction did you move?
- In which direction did the wave travel?

Conclude:

In a transverse wave, the oscillators move *perpendicular* to the direction in which the wave travels.

Part 2: Longitudinal Wave

Ask the students to stay in a line, looking at their neighbour, more distance between them Explain the new rules:

- Each student still represents one oscillator.
- Stay in your place.
- Watch the person before you.
- When they take one small step forward (or lean forward), wait about one second.
- Then do the same.
- After another second, step back to your original position.

Again, start with the first student.

Ask the students:

- Did anyone travel along the line?
- In which direction did you move?
- In which direction did the wave travel?
- How is this different from the previous activity?

Conclude:

In a longitudinal wave, the oscillators move *parallel* to the direction in which the wave travels.

Discussion

Compare the two activities together.

Transverse Wave	**Longitudinal Wave**
Oscillators move up and down.	Oscillators move back and forth.
Motion is perpendicular to the wave direction.	Motion is parallel to the wave direction.
Example: wave on a rope, guitar string.	Example: sound wave, slinky compression wave.

Finish with the key idea:

An oscillator moves back and forth around one position.

A wave is created when many oscillators pass this motion on to their neighbours.

*Experiment: Discovering a Harmonic Wave
Materials:

- One slinky per group (or one large slinky for the whole class)
- Small coloured stickers or pieces of tape

Place a few coloured stickers on different coils of the slinky so that the students can easily follow individual coils.

Part 1: Observe a Single Oscillator
Generate a slow, smooth transverse wave.
Ask the students:

Don't watch the whole slinky. Only watch the red coil.

Give them some time to observe.
Then ask:

- What is the red coil doing?
- Does it travel from one end of the slinky to the other?
- How does it move?
- Is the motion repeated?
- Does this motion remind you of something we have already studied?

Students should recognise that the coil moves just like a harmonic oscillator.
Write on the board:

$$x(t) = A\sin(2\pi ft) \tag{1}$$

Repeat the experiment while observing a different coloured coil.
Again ask:

- Does the blue coil move differently?
- Is it also performing a harmonic oscillation?

Guide the discussion towards the conclusion:

Every single coil behaves like a harmonic oscillator.The only difference is that they start at slightly different times. This time delay is called a phase shift.

$$x(t) = A\sin(2\pi ft + \varphi(x)) \tag{2}$$

Part 2: What Does the Whole Slinky Do?

Now ask the students to stop looking at a single coil and observe the entire slinky.
Ask:

- If every coil performs a harmonic oscillation, what does the whole slinky create?

Give the students time to discuss their ideas before introducing the term.
Finally conclude:

A **harmonic wave** is a wave in which every point of the medium performs a harmonic oscillation. At any instant, the whole wave has the shape of a sine curve. This is why we call it a harmonic wave.

Part 3: Group discussions

Discussion

1. **Which diagram could be used to describe the movement of one particular oscillator?**

   Answer: A graph of **displacement versus time**, $y(t)$.

   It shows how one single oscillator moves back and forth. For a harmonic oscillator, this motion is described by a sine or cosine function.

2. **Which diagram could be used to describe the appearance of the wave at one particular instant? What happens when time passes?**

   Answer: A graph of **displacement versus position**, $y(x)$.

   At one instant, the wave has the shape of a sine curve. As time passes, this sine curve moves through space while keeping its shape (for an ideal harmonic wave).

3. **Do we need additional quantities to describe a wave?**

   Answer: Yes. Besides the amplitude A and the frequency f, we also need

   - the **wavelength** λ, which describes the distance between two neighbouring crests (or troughs),
   - the **wave speed** c, which tells us how fast the wave travels through the medium.

   These quantities are related by

   $c = f\lambda$.

4. **We already know how to describe the motion of one oscillator mathematically. How can we describe the entire wave?**

   Answer:

   A single oscillator is described by

   $y(t) = A\sin(\omega t)$.

   A wave depends on both **time** and **position**. Therefore, we need a function with two variables:

   $y(x,t) = A\sin(2\pi ft - kx)$.

   This equation is called the **wave equation**. It describes how the shape of the wave changes in both space and time.

https://physics.bu.edu/ duffy/classroom.html simulation

Extension: Harmonic vs. Non-Harmonic Waves - optional

Now compare two different motions.

Experiment A: Generate a slow, smooth and regular wave.
Ask:

- Is the motion of one coil regular?
- Would you describe it as harmonic?

Experiment B: Shake the slinky irregularly so that no clear pattern forms.
Again ask the students to observe only one coloured coil.
Questions:

- Is the motion still regular?
- Does it still look like a sine oscillation?
- Is this still a harmonic oscillator?

Discuss the differences together.

Smooth Wave	**Irregular Wave**
Regular oscillation	Irregular oscillation
Periodic motion	Non-periodic motion
Each coil behaves like a harmonic oscillator	The motion is no longer harmonic
Harmonic wave	Non-harmonic wave

Key Idea

Key Idea

A harmonic wave is created when

- every point of the medium is a harmonic oscillator,
- all oscillators perform the same oscillation,
- neighbouring oscillators are shifted slightly in time (or phase),
- the disturbance travels through the medium.

Discussion

Finish by asking the class:

- Why does the wave travel even though no coil travels from one end of the slinky to the other?
- What is moving: the material or the disturbance?
- Why do we call this a *harmonic* wave?

3 From Waves to Sound

The Big Question

Conclude the lesson by asking the students:

> *Today we learned what a wave is.*
>
> *Can you think of a wave that is always around us, even though we cannot see it?*

.

Possible answers include:

- Sound
- Water waves
- Light
- Radio waves

Now focus on sound.

What exactly is sound?

Experiment

Strike a tuning fork (or play a pure tone with a loudspeaker).
Ask the students:

- What is oscillating?
- Can you see the sound travelling?
- If the tuning fork is oscillating, how does the sound reach your ears?

It is not the tuning fork that is moving around, oscillations are moving as sound wave through the air (pressure gradient) - surrounding air oscillate.

Connecting to Today's Lesson

Draw the following picture on the board:

$$\boxed{\text{Tuning fork}} \qquad \circ \;\circ \;\circ \;\circ \;\circ \;\circ \;\circ$$

Now ask:

- What are the oscillators now?
- How are they connected?
- Which type of wave is this?

Discuss the answers.

Oscillators	Air molecules
Coupling	Pressure forces and collisions
Type of wave	Longitudinal wave
Energy source	Oscillating tuning fork

What Happens Inside the Air?

Ask the students:

> *If the air molecules only move back and forth, why does the sound travel through the room?*

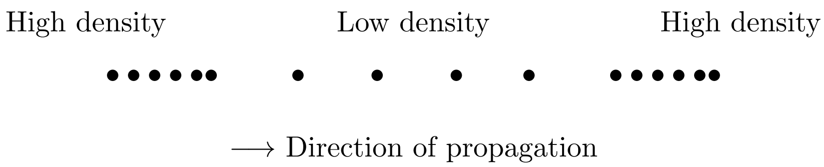
Discuss the idea that each air molecule pushes on its neighbours.
As a result, the air does not have the same density everywhere.
Instead, regions of **high density** and **low density** are created.

- **Compression:** many air molecules are close together. The density and pressure are higher.
- **Rarefaction:** the air molecules are farther apart. The density and pressure are lower.

Although each individual air molecule only oscillates around its equilibrium position, these regions of high and low density move through the air.
This travelling pattern of compressions and rarefactions is the sound wave.

The air molecules do **not** travel from the sound source to your ear.
Instead, regions of **high and low density** (high and low pressure) travel through the air, carrying energy.
This travelling pattern is called a **sound wave**.

A simple sketch:



Ask the students:

- What is moving?
- The air molecules or the regions of high and low density?

Conclude together:

Key Idea

Each air molecule only oscillates around its equilibrium position.
The regions of **high density** (compressions) and **low density** (rarefactions) travel through the air.
This travelling pattern carries the sound energy from the source to your ear.

Connecting Everything

Ask the students:

Can we now explain sound using everything we learned today?

Collect the ideas together.
Possible answers:

- A sound wave consists of many oscillating air molecules.
- Each air molecule oscillates around its equilibrium position.
- The molecules transfer their motion to neighbouring molecules.
- The disturbance travels through the air.
- Energy is transported, but the air itself does not travel across the room.

Final Summary

What is Sound?

Sound is a **longitudinal mechanical wave**.
The oscillators are the air molecules.
Neighbouring molecules are coupled through pressure forces and collisions.
Each molecule oscillates back and forth around its equilibrium position, while the disturbance travels through the air carrying energy.

Looking Ahead

Finish the lesson with one final question:

> *If sound is made of tiny oscillating air molecules...*

How can these tiny oscillations become the music we hear?

Tell the students:

This is exactly what we will discover in the next lesson.